Anqi Tan

Diagnostic Device Engineer · Point-of-Care Readers & Lateral Flow Tests

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Ph.D. candidate building portable readers for plasmon-enhanced lateral flow tests and validating them with human samples in lab and community settings.

EDUCATION

Ph.D. in Bioengineering, University of Illinois Urbana-Champaign Aug 2022 – Expected May 2027

Advisor: Prof. Brian T. Cunningham · Nanosensors Group

B.S. in Mechanical Engineering (dual degree), UIUC & Zhejiang University Aug 2018 – Jun 2022

EXPERIENCE

Graduate Research Assistant, Nanosensors Group, UIUC Sep 2022 – Present

- Built a **$87 pocket-size NIR fluorescence reader** end to end: 750 nm excitation and custom optical filtering, Raspberry Pi imaging hardware, and an **Android control app (Kotlin)** with automated T/C analysis in **20 s per strip**; image quality comparable to a lab ImageQuant scanner; first-author manuscript in preparation.
- Contribute to a **$4.87M DoD (PRMRP)** project developing a multiplexed IL-8/sTNFR-1 plasmonic-fluor LFA and handheld reader to classify sepsis/ARDS phenotypes at the bedside; lead the reader development.
- Developed a competitive plasmonic-fluor lateral flow assay for salivary cortisol with a **≤0.12 ng/mL** detection limit (antibody loading, blocking, and sample-preparation optimization); collected saliva from **153 adults**; accepted for an **oral presentation at APHA 2026**.
- Contributed to an ultrasensitive plasmon-enhanced LFA antigen test for *Neisseria gonorrhoeae* with WashU School of Medicine (*ACS Nano Medicine* 2026).

Undergraduate Research Leader, ZJU-UIUC Institute 2019 – 2021

- Led a national-level research project designing a re-localizable remote-center-of-motion mechanism for minimally invasive surgery (**first-author paper, IEEE ICARM 2021**).

SELECTED PROJECTS

CRISPR Mechanism Explainer (Ashby Prize Hackathon, NCSA) — 1st Place Oct 2025

- Two-person team; found label bias in the public benchmark, built a balanced validation set with experimental-condition context, and fine-tuned an LLM (Qwen3) to classify CRISPR screen outcomes and explain the mechanism; shipped a Gradio chat demo.

PUBLICATIONS

First author: IEEE ICARM 2021 paper · APHA 2026 oral presentation · pocket fluorescence reader manuscript (in preparation). **Co-author:** 4 journal papers (*Chemical Reviews*, *Lab on a Chip*, *ACS Nano Medicine*, *Biosensors*) and 1 Elsevier book chapter.

HONORS & AWARDS

1st Place, Ashby Prize Computational Science Hackathon, NCSA (2025) · Honorable Mention, Ashby Prize Hackathon (2024) · National Silver Medal, China "Internet+" Innovation & Entrepreneurship Competition (2021)

TEACHING

TA, BIOE/ECE 415 Biomedical Instrumentation Lab (Fall 2025) · TA, ME 170 Computer-Aided Design (3 semesters)

SKILLS

Engineering: SolidWorks, Creo, Fusion 360, Blender, 3D printing, laser cutting, optical/instrument design, COMSOL, FEA, design for assembly

Programming: Python, MATLAB, LabVIEW, Android app development (Kotlin), Raspberry Pi, Arduino, ESP32

Biosensing: lateral flow assays (competitive & sandwich), plasmonic-fluor labels, immunoassay optimization, ELISA, saliva/plasma sample prep, 3D cell culture, fluorescence imaging

Languages: English (fluent), Chinese (native)